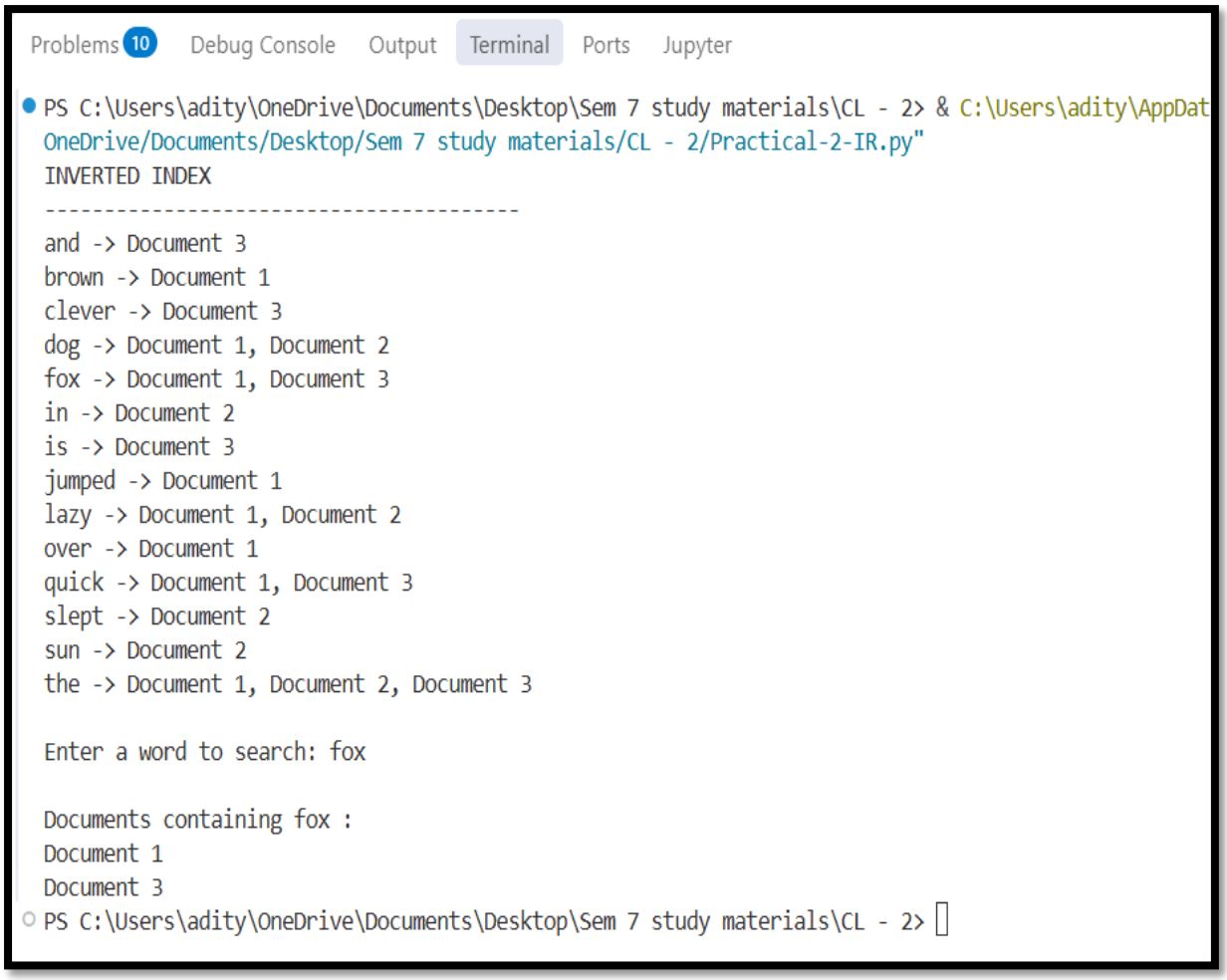
ASSIGNMENT No: 02 (IR)
Name: Aditya Rade
Roll.No:BEAD23167

Title: Implement a program for retrieval of documents using inverted files.
Code:

```
import re
documents={"Document 1":"The quick brown fox jumped over the lazy dog.","Document
2":"The lazy dog slept in the sun.","Document 3":"The fox is quick and clever."}
inverted_index={}
for document_id,text in documents.items():
   words=re.findall(r'\b\w+\b',text.lower())
   for word in words:
      if word not in inverted_index:
         inverted_index[word]=[]
      if document_id not in inverted_index[word]:
         inverted_index[word].append(document_id)
print("INVERTED INDEX")
print("-"*40)
for word in sorted(inverted_index):
   print(word,"->",", ".join(inverted_index[word]))
def retrieve_documents(query):
   return inverted_index.get(query.lower().strip(),[])
search_term=input("\nEnter a word to search: ")
result=retrieve_documents(search_term)
if result:
   print("\nDocuments containing",search_term,":")
   for document in result:
      print(document)
else:
   print("\nNo documents found for:",search_term)
print("Aditya Rade   BEAD23167")
```

==========

OUTPUT:


Problems 10
Debug Console
Output
Terminal
Ports
Jupyter
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppDat
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Practical-2-IR.py"
INVERTED INDEX
---
and -> Document 3
brown -> Document 1
clever -> Document 3
dog -> Document 1, Document 2
fox -> Document 1, Document 3
in -> Document 2
is -> Document 3
jumped -> Document 1
lazy -> Document 1, Document 2
over -> Document 1
quick -> Document 1, Document 3
slept -> Document 2
sun -> Document 2
the -> Document 1, Document 2, Document 3
Enter a word to search: fox
Documents containing fox :
Document 1
Document 3
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2>